

# **Data Management Assignment**

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**Word Count: 2,596 words**

## **1.0 Introduction**

Insurance organisations rely on high-quality data to support operational efficiency, risk management, and strategic decision-making (Davenport & Harris, 2007; Kimball & Ross, 2013). As customer portfolios grow and products diversify, data often exists across multiple systems and formats, making analysis complex and time-consuming (Inmon, 2005). The ability to integrate, clean, and analyse such data is therefore a critical capability for modern analytics teams (Batini & Scannapieco, 2016).

This assignment examines a scenario in which an insurance company plans to establish a dedicated analytics division. The immediate objective is to explore differences in customer characteristics across insurance products and preferred communication channels. The organisation faces challenges due to fragmented data and inconsistent quality, a common issue in legacy insurance systems (McKinsey Global Institute, 2016). While analysts are proficient in SQL, the company intends to gradually transition to R for advanced analytics and is evaluating big data technologies to handle growing volumes efficiently (Wickham & Grolemund, 2017).

The primary aim is to design a relational database, create an analytics base table (ABT), perform initial SQL-based descriptive analysis, conduct data quality assessment and cleaning in R, and generate actionable insights (Kimball & Ross, 2013). Microsoft Access was used for database design due to its relational capabilities, and R was used for data wrangling, cleaning, and visualisation (Kabacoff, 2022).

The report is structured into four sections: database development and structure, data quality analysis, insights and recommendations, and big data technology evaluation. Appendices contain all SQL and R code used.

## **2.0 Database Documentation**

### **2.1 Database Development in Microsoft Access**

The database development process began with reviewing data dictionaries and Excel files (Coronel & Morris, 2019). Four Excel files representing different insurance entities—Customer, Motor Policies, Health Policies, and Travel Policies—were imported as individual tables to preserve original structure and ensure traceability of the source data (Date, 2019).

Primary keys were defined for each table: CustomerID, MotorID, HealthID, and TravelID. The Customer table served as the central table and contained foreign keys linking to policy tables. Relationships were defined in Access to ensure referential integrity, enabling queries across tables efficiently and preventing orphan records (Elmasri & Navathe, 2016).

An analytics base table (ABT) was created using SQL to consolidate customer demographics and insurance policy details into a single dataset. This flat structure is essential for downstream analysis, as most analytical techniques require a customer-level dataset rather than a highly normalised schema (Kimball & Ross, 2013).

## **2.2 Database Structure and Table Overview**

- **Customer Table:** Contains title, name, gender, age, occupation, location, preferred communication channel, and credit card type, along with foreign keys linking to insurance policies (Coronel & Morris, 2019).
- **Motor Policies:** Stores policy dates, type, vehicle value, exposure, claims, vehicle body type, and age category. These variables are critical for understanding insurance risk and claims behaviour (Werner & Modlin, 2016).
- **Health Policies:** Includes policy duration, coverage level, and number of dependents, which provide insights into household structure and health risk exposure (OECD, 2019).
- **Travel Policies:** Captures policy dates and insurance type, useful for identifying cross-selling opportunities and short-term risk profiles (McKinsey & Company, 2020).

This relational design ensures flexibility, integrity, and scalability, supporting both operational and analytical requirements (Elmasri & Navathe, 2016).

## **2.3 Table Relationships and Join Strategy**

Relationships were implemented using primary and foreign keys, linking the Customer table to each policy table. Left joins were used when constructing the ABT to retain all customers, including those without certain policies. Inner joins were avoided to prevent bias by excluding customers without products, which could distort customer-level analysis (Kimball & Ross, 2013). This approach ensures comprehensive analysis while preserving policy-specific information.

## **2.4 Initial SQL Analysis**

Initial descriptive analysis in SQL examined customer counts by communication channel, average vehicle values by motor policy type, claim distributions, and product combinations across insurance types (Date, 2019). These queries revealed useful insights but also highlighted data quality concerns, including missing values and implausible entries such as zero vehicle values and extreme ages, which are common in raw operational data (Batini & Scannapieco, 2016).

## **2.5 Data Quality Issues Identified**

The SQL analysis identified key quality issues:

- Missing policy-related values, particularly for health and travel insurance.
- Implausible age values, including negative and extreme ages.
- Zero vehicle values in motor policies.
- Exposure values outside the valid 0–1 range.

If left unresolved, these issues could distort analysis, making structured data quality assessment and cleaning in R essential for reliable analytics outcomes (Batini & Scannapieco, 2016).

## **3.0 Data Quality Analysis and Preparation**

High-quality data is fundamental to accurate analytics, particularly in insurance, where decisions influence pricing, risk assessment, customer targeting, and profitability (Davenport & Harris, 2007; Werner & Modlin, 2016). Following initial SQL analysis, detailed data quality assessment was conducted in R to address inconsistencies and implausible values (Kabacoff, 2022).

### **3.1 Importance of Data Quality**

Poor data quality can lead to inaccurate risk segmentation, misleading customer insights, and ineffective marketing strategies (Batini & Scannapieco, 2016). Age, vehicle value, exposure, and claims history directly influence underwriting and premium calculations in insurance analytics (Werner & Modlin, 2016). Therefore, a structured, transparent data quality process was critical before deeper analysis (Kimball & Ross, 2013).

### 3.2 Identification of Issues

Using R, exploratory analysis revealed:

- Missing values in policy-related variables, reflecting absence of products rather than collection errors (Little & Rubin, 2019).
- Unrealistic ages, including negative and values above 100, which are not plausible in real-world insurance datasets (Batini & Scannapieco, 2016).
- Motor insurance records with zero vehicle values, which undermine asset valuation and pricing accuracy (Werner & Modlin, 2016).
- Exposure values outside the expected 0–1 range, violating insurance exposure definitions (Ohlsson & Johansson, 2010).

### 3.3 Why Missing Data Matters

Missing values can bias results if treated incorrectly (Little & Rubin, 2019). Policy-related gaps often indicate no coverage rather than data loss. For example, excluding customers without health insurance would misrepresent the customer base and skew analysis toward high-value clients, reducing representativeness and analytical validity (Batini & Scannapieco, 2016).

### 3.4 Rationale for Filtering

Filtering implausible records was preferred over imputation (Little & Rubin, 2019). Imputation for unrealistic ages or zero vehicle values introduces artificial data, misrepresenting risk exposure and violating insurance business logic (Werner & Modlin, 2016). Filtering ensures that retained data reflects genuine, interpretable customer information, supporting reliable descriptive analysis (Kimball & Ross, 2013).

### 3.5 Data Cleaning Implementation

Data cleaning steps included:

- Standardising gender entries to ensure categorical consistency (Kabacoff, 2022).
- Restricting age to 0–100 and removing invalid records based on demographic plausibility (Batini & Scannapieco, 2016).
- Removing motor policies with zero vehicle values to preserve asset valuation integrity (Werner & Modlin, 2016).
- Validating exposure ranges according to insurance exposure definitions (Ohlsson & Johansson, 2010).

- Replacing missing categorical values (e.g., card type, occupation, policy type) with labels like “Unknown” or “No Policy” to preserve records while maintaining interpretability (Little & Rubin, 2019).

The cleaned ABT was saved for subsequent analysis.

### **3.6 Business Implications**

Poor data quality could lead to incorrect conclusions about demographics, product uptake, and claims behaviour (Davenport & Harris, 2007). Unrealistic ages distort marketing strategies; zero vehicle values underestimate asset coverage and pricing assumptions (Werner & Modlin, 2016). These errors may result in inappropriate pricing, ineffective campaigns, and increased operational and regulatory risks (McKinsey Global Institute, 2016).

### **3.7 Preventing Future Issues**

Recommendations include:

- Implementing validation rules at data entry (age ranges, vehicle values, exposure) to prevent errors upstream (Batini & Scannapieco, 2016).
- Clarifying missing value definitions (“unknown” vs “not applicable”) to improve analytical transparency (Little & Rubin, 2019).
- Conducting routine audits using R or SQL to ensure long-term data reliability and governance compliance (Kimball & Ross, 2013).

## **4.0 Insights, Analysis, and Business Recommendations**

Following the data cleaning process, the final analytics base table provided a reliable foundation for deeper descriptive analysis (Kimball & Ross, 2013). Using R and ggplot2, customer characteristics were examined across insurance products and preferred communication channels to identify meaningful patterns with direct business relevance (Kabacoff, 2022).

### **4.1 Customer Demographic Patterns**

Analysis of gender distribution revealed a relatively balanced customer base, indicating that the company’s insurance offerings appeal broadly across genders. This balance suggests that marketing strategies should avoid overly gender-specific messaging and instead focus on needs-based segmentation, which has been shown to be more effective in insurance marketing contexts (Kotler & Keller, 2016).

Age analysis revealed distinct patterns across insurance products. Younger customers were more likely to hold travel insurance, reflecting higher mobility and travel frequency (UNWTO, 2020). In contrast, older customers were more commonly associated with motor and health insurance products, aligning with asset ownership and healthcare needs (Werner & Modlin, 2016). These patterns highlight the importance of age-sensitive product positioning in insurance portfolios (Kotler & Keller, 2016).

#### **4.2 Location-Based Insights**

Geographic analysis showed meaningful differences between customers in urban and rural locations. Urban customers were more likely to prefer digital communication channels such as email, while rural customers showed stronger preferences for traditional channels. Prior research indicates that access to digital infrastructure and lifestyle differences significantly influence communication preferences (OECD, 2021). These differences suggest that a uniform communication strategy would be inefficient and that location-specific outreach could improve engagement and response rates (Chaffey & Ellis-Chadwick, 2019).

#### **4.3 Communication Channel Preferences**

The dominance of certain communication channels reflects both demographic factors and customer convenience (Kotler & Keller, 2016). Email emerged as a widely preferred channel due to its low cost, speed, and accessibility, consistent with findings in digital marketing research (Chaffey & Ellis-Chadwick, 2019). However, the continued relevance of other channels indicates that a multi-channel strategy remains essential. Over-reliance on a single channel could alienate specific customer segments, reducing overall campaign effectiveness (Verhoef et al., 2015).

#### **4.4 Product and Channel Alignment**

Clear alignment was observed between certain insurance products and communication channels. Customers with motor insurance were more responsive to direct and digital communication, while health insurance customers showed mixed preferences, possibly reflecting privacy concerns associated with sensitive health information (Acquisti, Brandimarte & Loewenstein, 2015). Travel insurance customers demonstrated strong engagement with digital channels, suggesting opportunities for time-sensitive promotional campaigns and personalised offers (Chaffey & Ellis-Chadwick, 2019).

#### **4.5 Claims Behaviour and Risk Implications**

Claims analysis showed that most customers had not made claims, which is typical in insurance portfolios and critical for profitability (Ohlsson & Johansson, 2010). Customers with claims tended to have higher vehicle values and longer exposure periods. These insights can inform risk-adjusted pricing and targeted retention strategies, supporting more accurate underwriting and improved loss ratios (Werner & Modlin, 2016).

#### **4.6 Practical Marketing Recommendations**

Based on these findings, the company should adopt segmented marketing strategies that align products, demographics, and communication preferences (Kotler & Keller, 2016). Age-based targeting can improve message relevance, while location-specific channel selection can enhance response rates. Cross-selling opportunities exist for customers holding only motor insurance, particularly for travel and health products, which aligns with established customer lifetime value optimisation strategies in insurance marketing (Verhoef et al., 2015).

#### **4.7 Limitations and Future Opportunities**

This analysis is descriptive and does not establish causality. Future work could incorporate predictive modelling, behavioural data, and longitudinal analysis to enhance customer segmentation and lifetime value estimation, consistent with advanced analytics best practices (Davenport & Harris, 2007).

### **5.0 Big Data Technology for Future Analytics**

As the insurance company expands its customer base and diversifies its product offerings, both the volume and complexity of data are expected to grow exponentially (McKinsey Global Institute, 2016). Current tools, including Microsoft Access and standalone R environments, are well-suited for small to medium datasets but are limited in handling large-scale, heterogeneous, and rapidly evolving data sources (Chen, Mao & Liu, 2014). These traditional tools struggle with performance, parallel processing, real-time analytics, and integration of diverse data types, creating bottlenecks in operational and strategic decision-making. Therefore, adopting a scalable and robust big data technology is essential to ensure continued analytical efficiency and competitive advantage (Marz & Warren, 2015).

After careful evaluation, Apache Spark is recommended as the most suitable technology for the company's analytics roadmap. Spark is a distributed computing framework that processes

large-scale data efficiently by partitioning workloads across multiple nodes in a cluster (Zaharia et al., 2016). Its in-memory computation model allows iterative operations, such as aggregations, joins, and predictive analytics, to execute significantly faster than disk-based processing frameworks. Spark is also compatible with a variety of data sources, including relational databases, NoSQL stores, cloud data lakes, streaming systems, and external APIs, making it highly adaptable for the company's future data ecosystem (Karau et al., 2015).

Justification for Spark Selection is grounded in several factors. First, the company's current analytics team is proficient in SQL and is transitioning toward R for advanced analysis. Spark's dual support through Spark SQL and SparkR allows the team to leverage existing skills, facilitating a smooth adoption curve without extensive retraining (Karau et al., 2015). Second, Spark can handle both batch and real-time data processing, enabling the company to integrate streaming customer interactions, IoT-based telematics, and external risk indicators into the analytics pipeline (Zaharia et al., 2016). This capability is crucial for dynamic risk assessment, fraud detection, and personalized marketing strategies (McKinsey Global Institute, 2016). Third, Spark's modular architecture ensures future scalability: as the company grows, additional nodes can be added to the cluster to maintain high performance.

Implementation Steps would involve several stages:

1. **Infrastructure Setup:** Deploy a Spark cluster on-premises or via a cloud provider with sufficient compute and memory resources (Marz & Warren, 2015).
2. **Data Ingestion and Integration:** Connect Spark to existing Microsoft Access databases, cleaned R datasets, and external sources using distributed data ingestion pipelines (Karau et al., 2015).
3. **Analytics Pipeline Design:** Develop ETL workflows for cleaning, aggregation, and enrichment using Spark DataFrames (Zaharia et al., 2016).
4. **Workflow Automation and Scheduling:** Schedule recurring jobs to refresh analytics base tables (Marz & Warren, 2015).
5. **Monitoring and Governance:** Implement access controls, logging, and compliance monitoring aligned with insurance regulations (OECD, 2021).

Scalability and Performance Considerations highlight Spark's ability to support horizontal scaling and real-time analytics, enabling faster insights and improved operational agility (Zaharia et al., 2016).

Benefits include faster processing, advanced analytics, integration flexibility, and future-proof scalability, while challenges include infrastructure cost, training requirements, and governance complexity, all of which can be mitigated through phased adoption and governance frameworks (Karau et al., 2015; McKinsey Global Institute, 2016).

In conclusion, Apache Spark represents a scalable, flexible, and strategically aligned solution for the company's future analytics needs.

## **6.0 Conclusion**

This assignment demonstrated an end-to-end data management and analytics process that closely reflects real-world practices in the insurance industry (Davenport & Harris, 2007). Beginning with fragmented and heterogeneous Excel datasets, a structured relational database was designed and implemented in Microsoft Access. Clear identification of primary and foreign keys, along with the use of left joins, ensured data integrity and enabled the construction of a comprehensive analytics base table using SQL (Kimball & Ross, 2013). Initial descriptive analysis provided valuable high-level insights while also exposing several critical data quality issues.

A detailed data quality assessment and cleaning process was then undertaken using R. By applying transparent and business-justified rules such as removing implausible values and clearly labelling missing categorical data, the final dataset was both reliable and representative of the full customer base (Batini & Scannapieco, 2016). This clean dataset enabled meaningful analysis of customer demographics, insurance product ownership, communication preferences, and claims behaviour.

The insights generated support practical marketing and customer engagement strategies, including improved channel selection and targeted cross-selling opportunities (Kotler & Keller, 2016). Finally, the evaluation of Apache Spark highlighted how the organisation can scale its analytics capabilities to meet future demands (Zaharia et al., 2016). Overall, the assignment underscores the importance of robust data foundations in enabling effective, scalable, and insight-driven decision-making in modern insurance analytics.

## **Appendix 1: SQL Code (Microsoft Access)**

This appendix presents the SQL code used in Microsoft Access to construct the Analytics Base Table (ABT) and to perform initial descriptive analysis. These queries were used before any data quality issues were addressed, in line with the assignment requirements.

### **A1.1 Analytics Base Table (ABT) Query**

SELECT

customer.CustomerID,  
customer.Gender,  
customer.Age,  
customer.Location,  
customer.ComChannel,  
motor\_policies.MotorType,  
motor\_policies.veh\_value,  
motor\_policies.Exposure,  
motor\_policies.clm,  
motor\_policies.Numclaims,  
health\_policies.HealthType,  
travel\_policies.TravelType

FROM

(  
(  
customer  
LEFT JOIN motor\_policies  
ON customer.MotorID = motor\_policies.MotorID  
)  
)

```
LEFT JOIN health_policies
    ON customer.HealthID = health_policies.HealthID
)
LEFT JOIN travel_policies
    ON customer.TravelID = travel_policies.TravelID;
```

This query was saved in Microsoft Access as **ABT\_QUERY** and forms the basis for all subsequent SQL analysis.

### **A1.2 SQL Query 1: Customers by Preferred Communication Channel**

```
SELECT
    ComChannel,
    COUNT(CustomerID) AS TotalCustomers
FROM
    ABT_QUERY
GROUP BY
    ComChannel;
```

### **A1.3 SQL Query 2: Average Vehicle Value by Motor Policy Type**

```
SELECT
    MotorType,
    AVG(veh_value) AS AvgVehicleValue
FROM
    ABT_QUERY
GROUP BY
    MotorType;
```

### **A1.4 SQL Query 3: Claims Distribution**

```
SELECT
```

```
    clm,  
  
    COUNT(CustomerID) AS CustomerCount  
  
FROM  
  
    ABT_QUERY  
  
GROUP BY  
  
    clm;
```

#### **A1.5 SQL Query 4: Product Combination Analysis**

```
SELECT  
  
    MotorType,  
  
    HealthType,  
  
    TravelType,  
  
    COUNT(CustomerID) AS CustomerCount  
  
FROM  
  
    ABT_QUERY  
  
GROUP BY  
  
    MotorType,  
  
    HealthType,  
  
    TravelType;
```

### **Appendix 2: R Code**

This appendix includes all R code used for importing the original Excel files, creating the Analytics Base Table (ABT), conducting data quality checks, cleaning the data, performing descriptive analysis, and generating basic visualisations. The code is intentionally simple and well commented.

```
install.packages("readxl","dplyr","tidyr","ggplot2")
```

```
## Import the excel files
```

```
Data_1_Customer <- read_excel("C:/Sanjai/Data Management/assignment files/Data  
1_Customer.xlsx")
```

```
View(Data_1_Customer)
```

```
library(readxl)
```

```
Data_2_Motor_Policies <- read_excel("C:/Sanjai/Data Management/assignment files/Data  
2_Motor Policies.xlsx")
```

```
View(Data_2_Motor_Policies)
```

```
library(readxl)
```

```
Data_3_Health_Policies <- read_excel("C:/Sanjai/Data Management/assignment files/Data  
3_Health Policies.xlsx")
```

```
View(Data_3_Health_Policies)
```

```
library(readxl)
```

```
Data_4_Travel_Policies <- read_excel("C:/Sanjai/Data Management/assignment files/Data  
4_Travel Policies.xlsx")
```

```
View(Data_4_Travel_Policies)
```

```
## Load Libraries
```

```
library(dplyr)
```

```
library(tidyr)
```

```
library(ggplot2)
```

```
## Join Files to Create ABT in R
```

```
abt <- Data_1_Customer %>%
```

```
  left_join(Data_2_Motor_Policies, by = "MotorID") %>%
```

```
  left_join(Data_3_Health_Policies, by = "HealthID") %>%
```

```
  left_join(Data_4_Travel_Policies, by = "TravelID")
```

```
## Summarise the data

str(abt)

summary(abt)

## Data Quality Analysis

## Check the missing values

colSums(is.na(abt))


## Outliers

summary(abt$Age)

summary(abt$veh_value)

## Visual check

boxplot(abt$Age)

boxplot(abt$veh_value)

## Data Cleaning

## Fix Gender values

abt$Gender <- toupper(abt$Gender)

## Remove unrealistic ages

abt <- abt %>% filter(Age >= 0 & Age <= 100)

## Remove zero vehicle values

abt <- abt %>% filter(veh_value > 0)

## Replace missing card type

abt$CardType[is.na(abt$CardType)] <- "Unknown"

## Exposure basic validation

abt <- abt %>% filter(Exposure >= 0 & Exposure <= 1)
```

```
## Visual checks

## Age distribution

ggplot(abt, aes(x = Age)) +

  geom_histogram(binwidth = 5)

## Vehicle value distribution

ggplot(abt, aes(x = veh_value)) +

  geom_histogram(binwidth = 1)

## Summarise the data

summary(abt)

colSums(is.na(abt))

## Fix occupation missing

abt$Occupation[is.na(abt$Occupation)] <- "Unknown"

## Fix HealthType missing

abt$HealthType[is.na(abt$HealthType)] <- "No Health Policy"

## Fix TravelType missing

abt$TravelType[is.na(abt$TravelType)] <- "No Travel Policy"

## Summaries

## Customers by Gender

table(abt$Gender)

## Customers by Location

table(abt$Location)

## Average age

mean(abt$Age)

## Claims distribution
```

```
table(abt$clm)
```

```
## Visualisation
```

```
## This plot shows how customers are distributed across different gender categories
```

```
ggplot(abt, aes(x = Gender)) +
```

```
  geom_bar() +
```

```
  ggtitle("Distribution of Customers by Gender")
```

```
## This plot compares the number of customers living in Urban and Rural locations
```

```
ggplot(abt, aes(x = Location)) +
```

```
  geom_bar() +
```

```
  ggtitle("Distribution of Customers by Location")
```

```
## Save final cleaned dataset
```

```
write.csv(abt, "abt_clean.csv", row.names = FALSE)
```

### Appendix 3: Additional Outputs

#### Summary: ABT

```
summary(abt)
```

```
CustomerID  Title      GivenName  MiddleInitial  Surname
Min.   : 1 Length:4085   Length:4085   Length:4085   Length:4085
1st Qu.:1294 Class :character Class :character Class :character Class :character
Median :2591 Mode  :character Mode  :character Mode  :character Mode  :character
Mean   :2601
3rd Qu.:3905
Max.   :5200
```

```
CardType      Occupation  Gender      Age      Location
Length:4085   Length:4085 Length:4085  Min.   :-44.00 Length:4085
```

Class :character Class :character Class :character 1st Qu.: 22.00 Class :character  
 Mode :character Mode :character Mode :character Median : 46.00 Mode :character  
 Mean : 41.38  
 3rd Qu.: 50.00  
 Max. :210.00

ComChannel MotorID HealthID TravelID  
 Length:4085 Min. :1004 Min. :1001 Min. :1001  
 Class :character 1st Qu.:3202 1st Qu.:3222 1st Qu.:3274  
 Mode :character Median :5593 Median :5498 Median :5489  
 Mean :5533 Mean :5509 Mean :5535  
 3rd Qu.:7773 3rd Qu.:7759 3rd Qu.:7872  
 Max. :9999 Max. :9999 Max. :9996  
 NA's :728 NA's :1547 NA's :1980

PolicyStart PolicyEnd.x MotorType  
 Min. :2019-01-01 00:00:00 Min. :2020-01-01 00:00:00 Length:4085  
 1st Qu.:2019-04-06 00:00:00 1st Qu.:2020-04-06 00:00:00 Class :character  
 Median :2019-07-04 00:00:00 Median :2020-07-04 00:00:00 Mode :character  
 Mean :2019-07-04 14:48:47 Mean :2020-07-04 11:12:36  
 3rd Qu.:2019-10-04 00:00:00 3rd Qu.:2020-10-04 00:00:00  
 Max. :2019-12-30 00:00:00 Max. :2020-12-30 00:00:00  
 NA's :728 NA's :728

veh\_value Exposure clm Numclaims v\_body  
 Min. : 0.000 Min. :0.00274 Min. :0.00000 Min. :0.00000 Length:4085  
 1st Qu.: 1.030 1st Qu.:0.21629 1st Qu.:0.00000 1st Qu.:0.00000 Class :character  
 Median : 1.510 Median :0.48460 Median :0.00000 Median :0.00000 Mode :character  
 Mean : 1.812 Mean :0.47804 Mean :0.06375 Mean :0.06822  
 3rd Qu.: 2.220 3rd Qu.:0.77207 3rd Qu.:0.00000 3rd Qu.:0.00000  
 Max. :16.690 Max. :0.99932 Max. :1.00000 Max. :3.00000  
 NA's :728 NA's :728 NA's :728 NA's :728

v\_age LastClaimDate policyStart.x  
 Min. :1.000 Min. :2019-02-02 00:00:00 Min. :2019-01-01 00:00:00

1st Qu.:2.000 1st Qu.:2019-09-28 00:00:00 1st Qu.:2019-03-29 00:00:00  
 Median :3.000 Median :2019-12-25 00:00:00 Median :2019-07-02 00:00:00  
 Mean :2.659 Mean :2019-12-25 14:21:18 Mean :2019-06-30 18:54:11  
 3rd Qu.:4.000 3rd Qu.:2020-04-04 00:00:00 3rd Qu.:2019-09-28 18:00:00  
 Max. :4.000 Max. :2020-12-06 00:00:00 Max. :2019-12-31 00:00:00  
 NA's :728 NA's :3871 NA's :1547  
 policyEnd HealthType HealthDependentsAdults DependentsKids  
 Min. :2020-01-01 00:00:00 Length:4085 Min. :0.0000 Min. : 0.000  
 1st Qu.:2020-03-29 00:00:00 Class :character 1st Qu.:0.0000 1st Qu.: 0.000  
 Median :2020-07-02 00:00:00 Mode :character Median :1.0000 Median : 2.000  
 Mean :2020-06-30 14:51:54 Mean :0.8164 Mean : 1.764  
 3rd Qu.:2020-09-28 18:00:00 3rd Qu.:1.0000 3rd Qu.: 3.000  
 Max. :2020-12-31 00:00:00 Max. :2.0000 Max. :40.000  
 NA's :1547 NA's :1547 NA's :1547  
 policyStart.y PolicyEnd.y TravelType  
 Min. :2020-01-01 00:00:00 Min. :2020-01-04 00:00:00 Length:4085  
 1st Qu.:2020-07-09 00:00:00 1st Qu.:2020-07-19 00:00:00 Class :character  
 Median :2020-07-29 00:00:00 Median :2020-08-10 00:00:00 Mode :character  
 Mean :2020-07-21 20:52:33 Mean :2020-07-31 22:07:07  
 3rd Qu.:2020-08-19 00:00:00 3rd Qu.:2020-08-31 00:00:00  
 Max. :2020-12-31 00:00:00 Max. :2021-01-09 00:00:00  
 NA's :1980 NA's :1980

## Data Quality Analysis

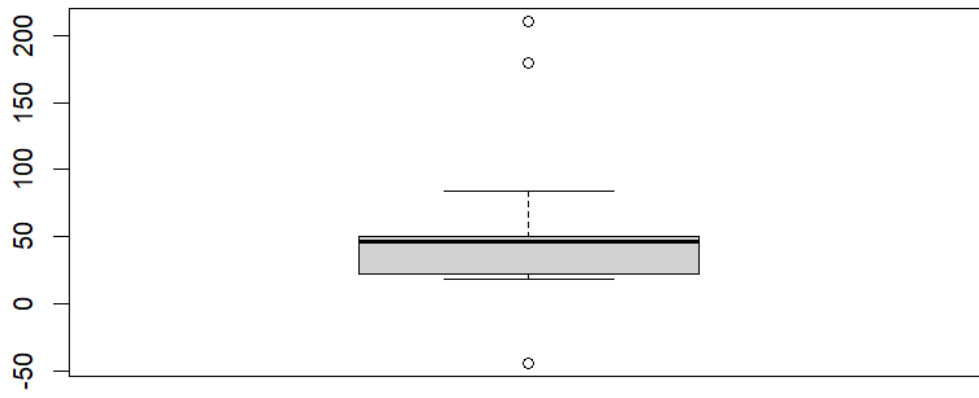
```
> colSums(is.na(abt))
```

| CustomerID | Title    | GivenName  | MiddleInitial |
|------------|----------|------------|---------------|
| 0          | 0        | 0          | 0             |
| Surname    | CardType | Occupation | Gender        |
| 0          | 0        | 1554       | 0             |
| Age        | Location | ComChannel | MotorID       |
| 0          | 0        | 0          | 728           |

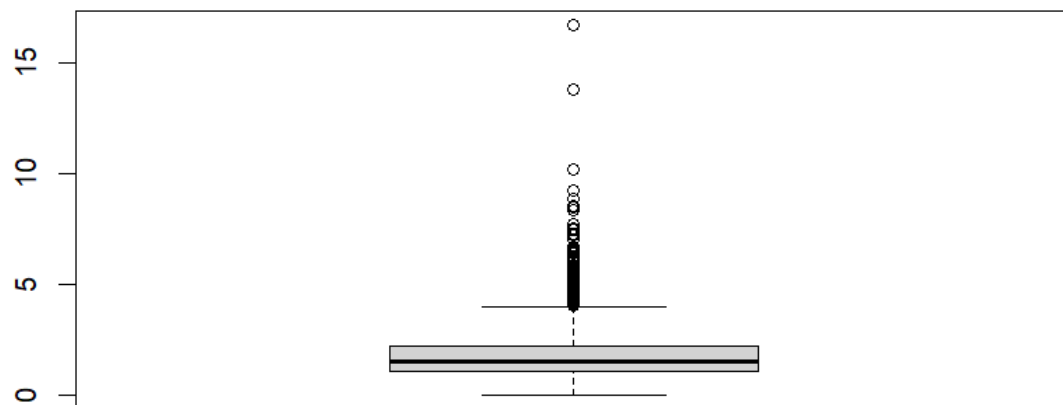
|                |               |             |                        |
|----------------|---------------|-------------|------------------------|
| HealthID       | TravelID      | PolicyStart | PolicyEnd.x            |
| 1547           | 1980          | 728         | 728                    |
| MotorType      | veh_value     | Exposure    | clm                    |
| 728            | 728           | 728         | 728                    |
| Numclaims      | v_body        | v_age       | LastClaimDate          |
| 728            | 728           | 728         | 3871                   |
| policyStart.x  | policyEnd     | HealthType  | HealthDependentsAdults |
| 1547           | 1547          | 1547        | 1547                   |
| DependentsKids | policyStart.y | PolicyEnd.y | TravelType             |
| 1547           | 1980          | 1980        | 1980                   |

**Boxplots :**

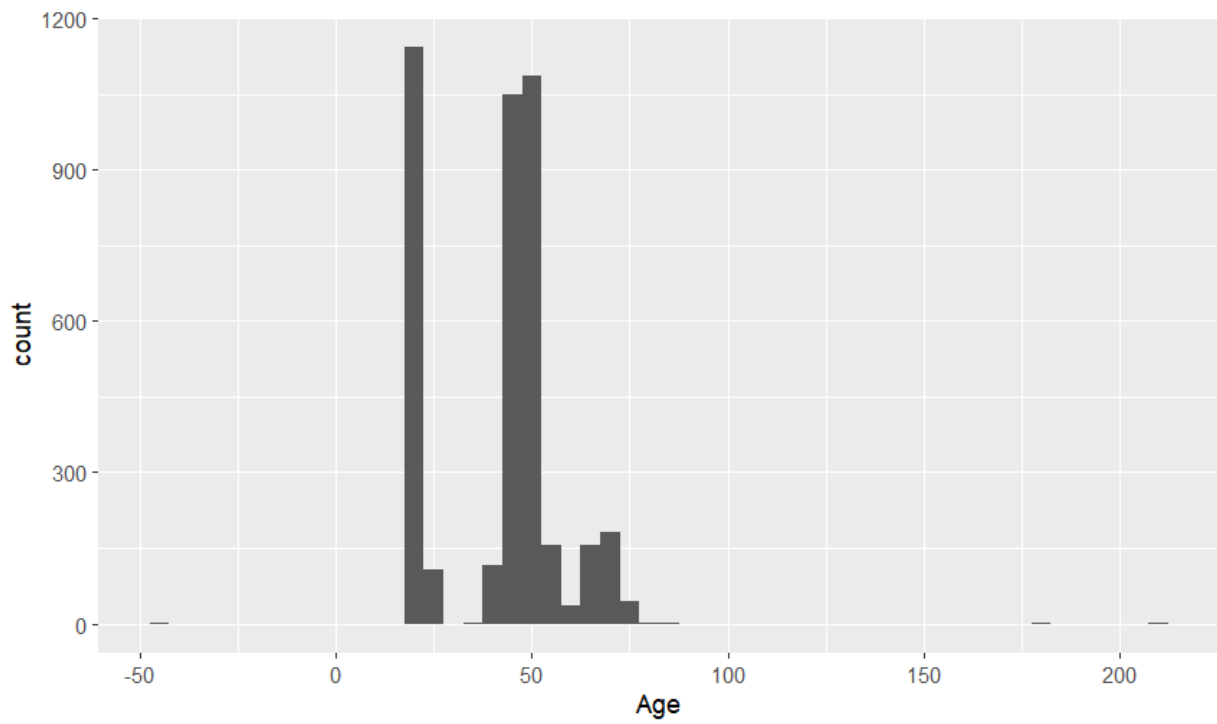
**abt\$age**



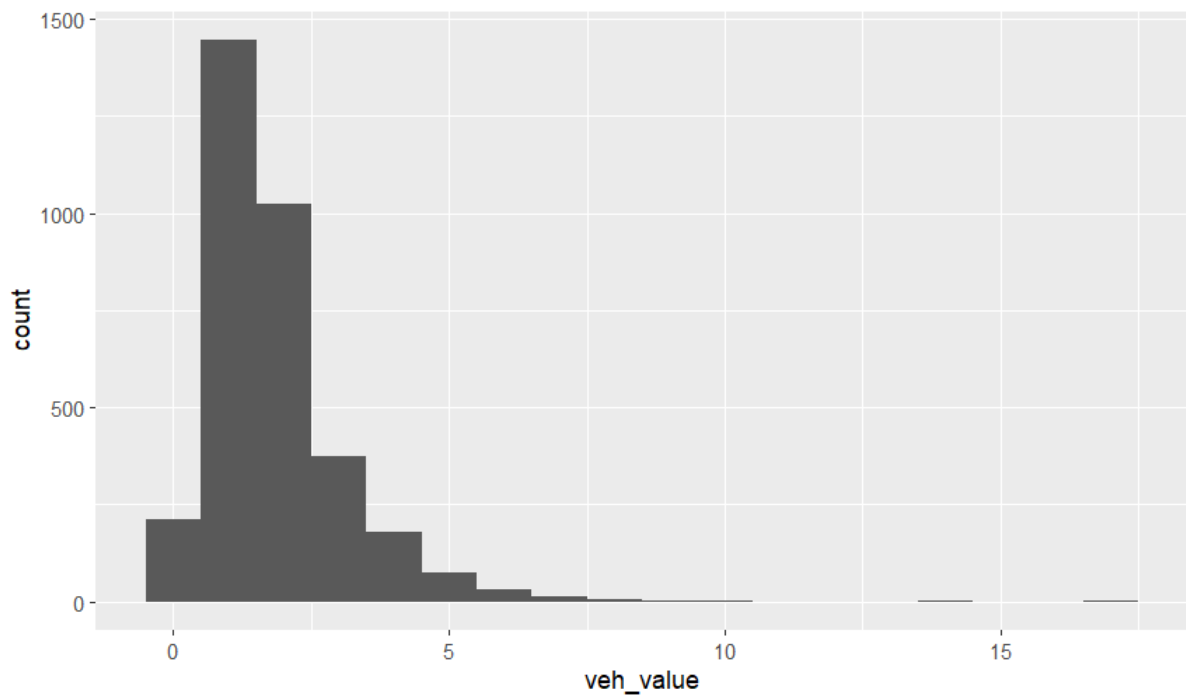
**abt\$veh\_value**



**Histogram: Age**



**Histogram: veh\_vslue**



## References

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